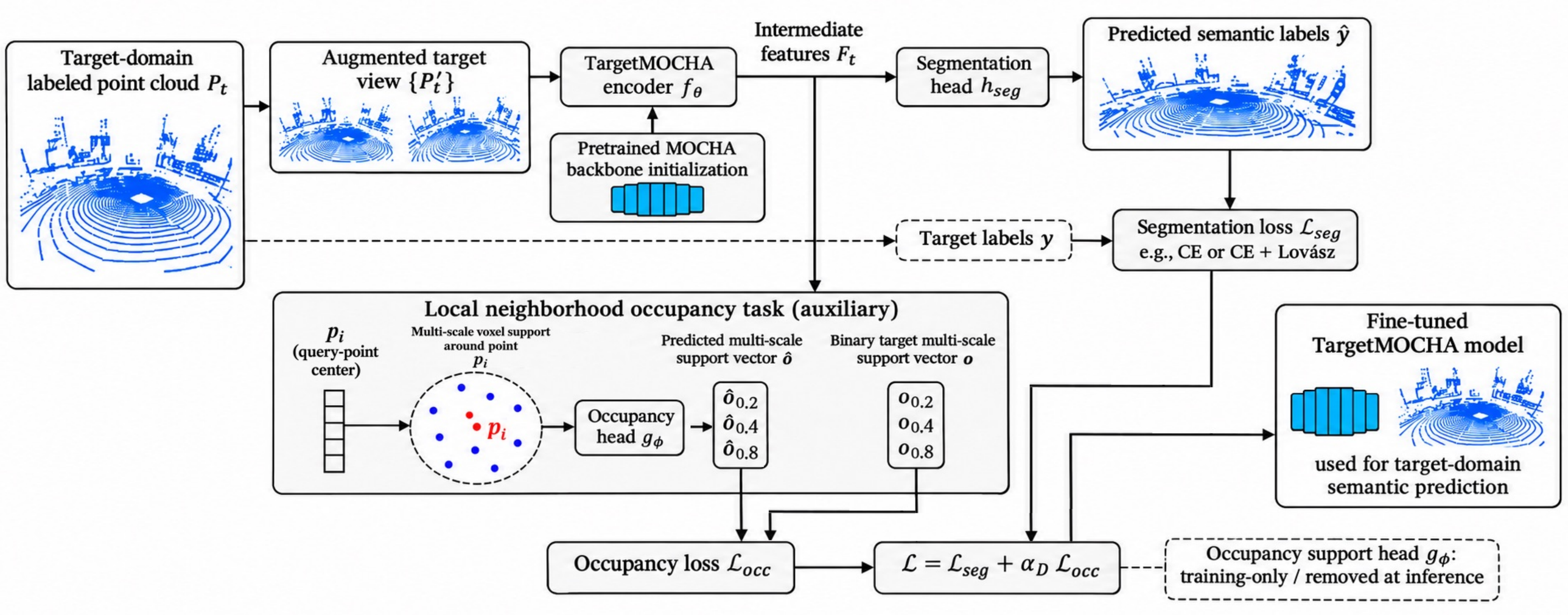

Target-domain labeled point cloud P_t
Augmented target view $\{P_t'\}$
TargetMOCHA encoder f_θ
Pretrained MOCHA backbone initialization
Intermediate features F_t
Segmentation head h_{seg}
Predicted semantic labels $\hat{y}$
Target labels y
Segmentation loss $\mathcal{L}_{seg}$
e.g., CE or CE + Lovász
Local neighborhood occupancy task (auxiliary)
p_i (query-point center)
Multi-scale voxel support around point p_i
Occupancy head g_ϕ
Predicted multi-scale support vector $\hat{o}$
$\hat{o}_{0.2}$ $\hat{o}_{0.4}$ $\hat{o}_{0.8}$
Binary target multi-scale support vector o
$o_{0.2}$ $o_{0.4}$ $o_{0.8}$
Occupancy loss $\mathcal{L}_{occ}$
$\mathcal{L} = \mathcal{L}_{seg} + \alpha_D \mathcal{L}_{occ}$
Occupancy support head g_ϕ: training-only / removed at inference
Fine-tuned TargetMOCHA model
used for target-domain semantic prediction